

SYLLABUS DIGITAL CIRCUITS & SYSTEMS

Course No	Title of the Course	Course Structure	Pre-Requisite
EAEPC305	Digital Circuits & Systems	3L-0T-2P	---
COURSE OUTCOMES (COs) After completion of this course, the students are expected to be able to demonstrate the following knowledge, skills and attitudes: CO 1: To get familiarized with number systems, codes, logic gates, logic families and Boolean algebra CO 2: To analyze and understand the design process associated with combinational Circuits. CO 3: To analyze and understand the design process associated with sequential circuit. CO 4: To understand fundamental concepts of VHDL modeling for basic digital circuits. CO 5: To develop basic understanding of programmable logic devices and convertors			
UNIT	CONTENTS		
1	Logic functions and operation of logic families: Digital design representations (e.g., truth table, equations, schematic, etc.), Different codes and their properties (e.g., Gray, BCD, Exess-3 etc.). Signed and unsigned Number representation and conversion. Binary addition and subtraction (two's complement). Boolean equations and terminology (SOP, POS, min-term, max-term), Logic implementations of equations (2-level AND-OR, NAND-NAND, NOR-NOR, etc.). Characteristics of TTL and CMOS logic families (i.e., logic thresholds, delay, Noise margin, fan-in and fan- out, power dissipation), TTL to CMOS and CMOS to TTL interfacing.		
2	Combinational logic: Overview of Boolean algebra, Simplification of logic functions using K-map up to 5 variables with don't care conditions, Tabulation method. Combinational logic design: half and full adder, half and full subtractor, BCD adder, Multiplexers, Demultiplexer, Encoders, Decoders, Priority encoders, Magnitude comparator. Code converters: Binary to Gray, BCD to Seven Segment Display etc, Introduction to a PROM and its use for implementing arbitrary logic functions.		
3	Sequential logic: SR Latch, D Latch, Master-Slave operation, Latch versus flip-flop, Flip-Flops (D, SR, JK, and T), Registers, Shift registers (SISO, SIPO, PISO), bidirectional shift-registers. Ripple and synchronous counters. Modulo-N counters. Synchronous and asynchronous sequential circuits. Mealy and Moore sequential models. Design of Finite State Machine using state table, state diagram, state assignment, next state and output equations. Example FSM implementations.		
4	Hardware Description Language: Modern digital design methods. Motivation and need for hardware description language. Introduction to VHDL and important terminology. VHDL Library and packages. Entities, architecture and configurations. Signals and data types. Operators and processes. VHDL code examples of various combinational and sequential circuits from unit-2 and unit-3.		
5	Digital System Design and Programmable Logic Devices: Transistor-level schematics of CMOS logic gates, Introduction to Digital to analog convertor, Analog to digital convertor, Digital system design using FSM Design approach. Elements of CPU-ALU, control logic and register file. Introduction to programmable logic devices - PAL/PLA, GAL, CPLD and FPGA.		
SUGGESTED READINGS 1. Donald D. Givone, "Digital Principles & Design", Tata McGraw Hill. 2. M.M. Mano, M.D. Ciletti, "Digital Design", 6th edition, Pearson, 2018. 3. C. Roth, L. Kinney, "Fundamentals of Logic Design", 7th Edition Cengage Learning, 2014. 4. R.P. Jain, "Modern Digital Electronics", McGraw Hill education, 4th Edition 5. Douglas L. Perry, "VHDL Programming by Examples", McGraw Hill, 4 th Edition. 6. J. Bhaskar, "VHDL Primer", 3 rd Edition, Pearson.			

Suggested List of experiments:

1. Design, Implement and verify a half-adder and full-adder,
 2. Design, Implement and verify a half-subtractor and full-subtractor
 3. Design, implement and verify a 2-bit binary comparator using basic gates.
 4. Design and verify Binary to Gray code converter.
 5. Design and verify Gray to Binary code converter.
 6. Design and verify 2 to 4 decoder.
 7. Design, Implement and verify a 3x8 decoder.
 8. Design, implement and verify 8x1 multiplexor.
 9. Design and implement J-K Flip/flop
 10. Design and verify SISO shift register
 11. Design, implement and verify a 4-bit binary counter and a decimal counter
 12. Design, implement and verify a MOD-12 up-counter.
- Note: Experiments may be simulated on Xilinx ISE Design Suite 14.7 or EDA Playground (cloud-based platform) for VHDL simulations.